

Comprehensive Guide to Line Search

Binary Search & Golden-Section Search

Detailed Study Guide for Part (e)

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1 Phase 0: The Setup (The Most Important Step)

Before you can perform *any* search, you must define the function you are searching over. You cannot just “plug numbers” into the algorithm without this equation.

1.1 Understanding the Goal

In Part (e), you have a current position w_{old} and a direction d . You want to take a step of size η (eta). The new position is:

$$w_{\text{new}} = w_{\text{old}} + \eta \cdot d$$

We want to find the specific η that makes the Loss Function $J(w_{\text{new}})$ as small as possible.

1.2 Constructing the Polynomial $\phi(\eta)$

The Transformation

We define a 1-dimensional function $\phi(\eta)$ by substituting the expression for w_{new} into J .

Step-by-Step Derivation Guide:

1. Identify Inputs:

- Let $w_{\text{old}} = [u, v]^T$ (Your values from the end of Part C).
- Let $d = [p, q]^T$ (Your descent direction vector, usually $-\nabla J$).

2. Parameterize the Weights: Write the components of the new weight vector in terms of η :

$$w_1(\eta) = u + \eta \cdot p$$

$$w_2(\eta) = v + \eta \cdot q$$

3. Substitute into Loss Function: If your loss function is, for example, $J(w_1, w_2) = (w_1 - 2)^2 + 3(w_2 - 4)^2$:

$$\phi(\eta) = ((u + \eta p) - 2)^2 + 3((v + \eta q) - 4)^2$$

4. Expand and Group (Crucial!): Do not leave it in parentheses. Expand the squares. Group all η^2 terms, all η terms, and constants. You will get a quadratic equation:

$$\phi(\eta) = A\eta^2 + B\eta + C$$

Use this simplified A, B, C form for all your calculations below.

2 Method 1: Binary Search

(Based on Slide 25: “Line search Algorithms - Binary search”)

2.1 The Logic

Binary search finds the minimum by determining which way the function is sloping.

- We pick the middle of the interval (m).
- We pick a tiny step to the right ($m + \epsilon$).
- We compare the heights to see if we are going uphill or downhill.

2.2 The Algorithm

Initialization: Interval $[a, b] = [0, 1]$. Small constant $\epsilon = 0.001$ (or as specified).

Calculate midpoint $m = \frac{a+b}{2}$. Calculate probe $m_\epsilon = \frac{a+b+\epsilon}{2}$.

1. **If $\phi(m) < \phi(m_\epsilon)$ (Increasing Slope):**

- The function is going UP to the right. The minimum is to the **LEFT**.
- **New Interval:** $[a, m_\epsilon]$
- *Note: The slide explicitly uses m_ϵ as the new upper bound.*

2. **If $\phi(m) > \phi(m_\epsilon)$ (Decreasing Slope):**

- The function is going DOWN to the right. The minimum is to the **RIGHT**.
- **New Interval:** $[m, b]$
- *Note: The slide explicitly uses m as the new lower bound.*

2.3 Detailed Practice Example

Task: Minimize $\phi(\eta) = (\eta - 0.4)^2 + 2$ on interval $[0, 1]$ with $\epsilon = 0.01$.

Step-by-Step Example: Iteration 1**1. Calculate Points:**

$$a = 0, \quad b = 1$$

$$m = \frac{0 + 1}{2} = 0.5$$

$$m_\epsilon = \frac{0 + 1 + 0.01}{2} = 0.505$$

2. Evaluate Function:

$$\phi(0.5) = (0.5 - 0.4)^2 + 2 = 0.01 + 2 = \mathbf{2.01}$$

$$\phi(0.505) = (0.505 - 0.4)^2 + 2 = 0.011025 + 2 = \mathbf{2.011025}$$

3. Compare:

$$2.01 < 2.011025 \implies \phi(m) < \phi(m_\epsilon)$$

The function is **Increasing**.

4. Update: We keep the Left side. New Interval: $[0, 0.505]$.

! Common Pitfall: Slope Calculation

Do **not** calculate the derivative $\phi'(\eta)$ to find the slope. The method requires you to compare the two values $\phi(m)$ and $\phi(m_\epsilon)$ directly.

3 Method 2: Golden-Section (Interval Elimination)

(Based on Slides 26–28: “Line search Algorithms - Golden-section search”)

3.1 The Logic

This method does not check the slope. Instead, it checks **three** zones using two internal points. It eliminates the zone that definitely does not contain the minimum.

Given interval $[a, b]$ and points m_1, m_2 (where $a < m_1 < m_2 < b$):

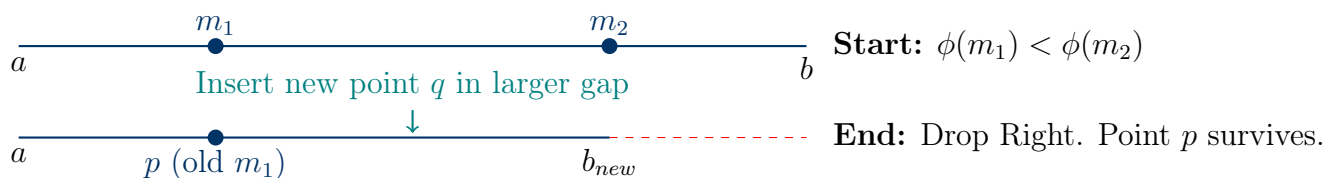
- **If $\phi(m_1) < \phi(m_2)$:** The minimum is closer to m_1 (Left).
- **Action:** Drop the segment $[m_2, b]$. New Interval: $[a, m_2]$.
- **If $\phi(m_1) > \phi(m_2)$:** The minimum is closer to m_2 (Right).
- **Action:** Drop the segment $[a, m_1]$. New Interval: $[m_1, b]$.

3.2 Slide 27: How to Pick the Next Point

Unlike standard Golden-Section search which uses a fixed ratio of 0.618, the slides describe a **General Interval Elimination** process.

1. After an update, you have a new interval and **one** remaining internal point (let's call it p).
2. You need a second point q .
3. **Rule:** Identify the two gaps inside the interval ($[a, p]$ and $[p, b]$). Place q inside the **larger** gap. (Usually at the midpoint of that gap).

3.3 Visualizing the Logic



3.4 Detailed Practice Example

Task: Minimize $\phi(\eta)$ on $[0, 1]$. Start with $m_1 = 0.25, m_2 = 0.75$.

Step-by-Step Example: Iteration 1

1. **Evaluate:** Assume $\phi(0.25) = 10$ and $\phi(0.75) = 20$.
2. **Compare:** $10 < 20$. The minimum is on the Left.
3. **Drop:** We discard the region right of m_2 . Drop $[0.75, 1]$.
4. **New State:** New Interval: $[0, 0.75]$. Remaining Point: $p = 0.25$.

Step-by-Step Example: Iteration 2 (Crucial Step)

1. **Analyze Gaps:** Current Interval: $[0, 0.75]$. Existing point: 0.25.
 - Gap A (Left): $[0, 0.25] \rightarrow$ Length 0.25.
 - Gap B (Right): $[0.25, 0.75] \rightarrow$ Length 0.50.
2. **Apply Slide Rule:** Gap B is larger. We must place the new point q in Gap B. A valid choice is the midpoint of Gap B:

$$q = \frac{0.25 + 0.75}{2} = 0.5$$

3. **New Points for Comparison:** We compare our existing point 0.25 and our new point 0.5. Evaluate $\phi(0.25)$ and $\phi(0.5)$ and repeat the Drop logic.

4 Stopping Criteria

When do you stop calculating? The slides mention “achieve the desired accuracy.”

1. **Interval Size:** Stop when the length of the interval $L = b - a$ is smaller than a threshold (e.g., $L < 0.01$). This guarantees the minimum is within that small range.
2. **Fixed Iterations:** For written assignments, you are often asked to perform exactly k iterations (e.g., 2 or 3). If specified, stop there.
3. **Gradient:** Stop when the difference $|\phi(m) - \phi(m_\epsilon)|$ is extremely close to zero.

5 Summary Worksheet for Part (e)

Use these tables to structure your solution for the assignment.

Binary Search Table

Iter	a	b	m	m_ϵ	Compare	New Interval
1	0	1	0.5	0.501	$\phi(m) \dots \phi(m_\epsilon)$	...
2	...	...	...	...	...	...

Golden-Section (Interval Elimination) Table

Iter	Interval	m₁	m₂	Values	Action	Next Point
1	[0, 1]	0.25	0.75	...	Drop ...	In larger gap
2	...	...	...	...	...	...